



Boosthub Staking v2

CurveYield Security

Smart-contract security audit

Source review, adversarial testing, pinned-fork verification, and remediation review

PASS

TECHNICAL STATUS

FINAL FINDING STATE

7 findings

C0 H0 M0 L0

DEPLOYMENT STATE UNATTESTED

05 SYSTEM OVERVIEW

Boosthub Staking v2 - audited architecture summary

BOOSTHUB STAKING V2

Boosthub Staking v2 was reviewed as a 8-file production smart-contract suite. Exact audited source identities and final versions are listed in Scope.

The architecture review covers custody, accounting, reward flows, privileged authority, external dependencies, and replacement/liveness constraints across that final source set.

8

contracts

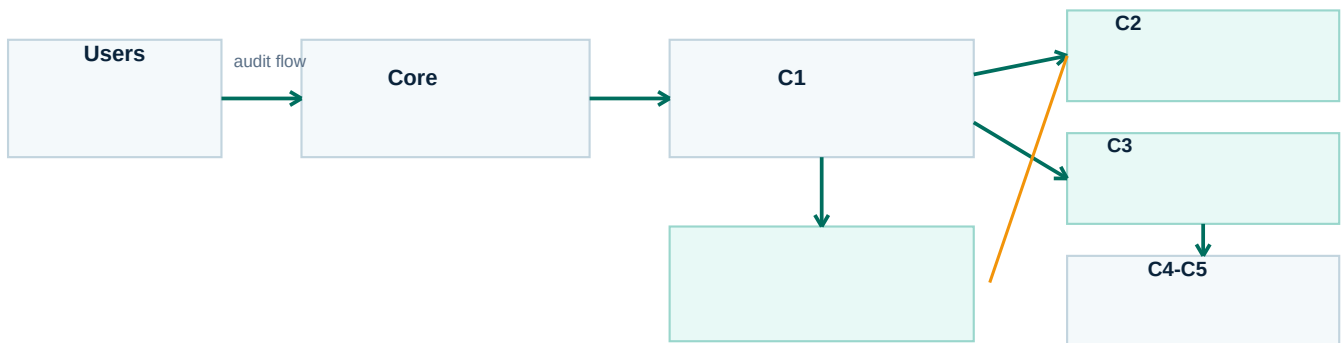
17

evidence files

AUDIT STATE

UNATTESTED

8 contracts in scope



AUDITED SYSTEM SUMMARY

1 P

The complete source-grounded System Overview is retained in the public overview provenance record.

2 U

The security review is bound to 8 declared smart contract(s) listed in Scope.

3 A

Contract behavior, findings, and evidence are evaluated against that declared source set.

4 R

Deployment attestation is reported separately from the smart-contract security verdict.

PUBLIC EVIDENCE

Evidence 1

Contract scope

Evidence 2

Contract findings

Evidence 3

Methods

02 TABLE OF CONTENTS

Report structure and finding locations

Overview 2

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Scope 5-7

Threats 8

Methods 9-13

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Claims 16

Fixes 19-27

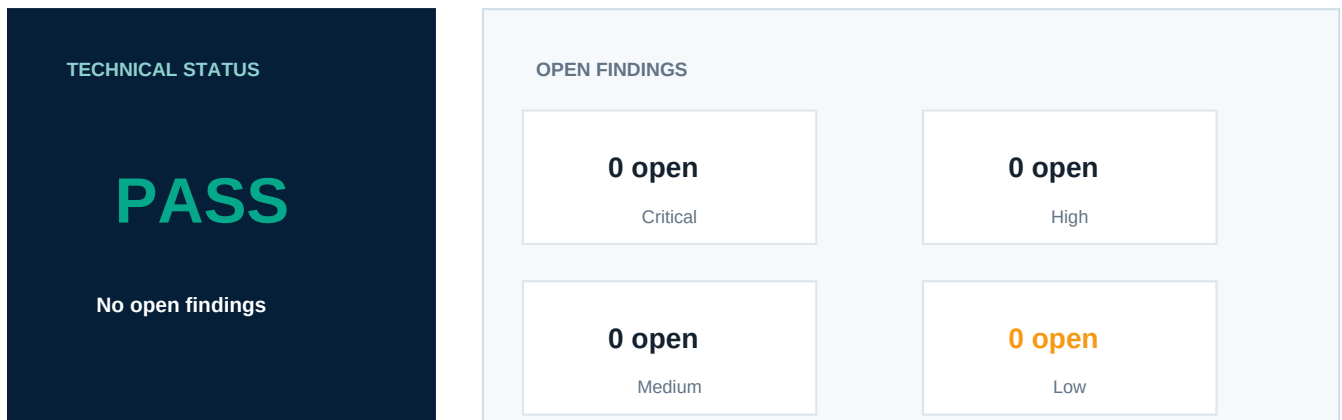
Findings 17-26

Evidence 28-30

End 31

03 EXECUTIVE SUMMARY

Final security position and the evidence that drove it



FINDING DISPOSITION

BH-V2-LOW-001 ACCEPT

BH-V2-LOW-002 FIXED

BH-V2-LOW-003 ACCEPT

ACPT

FIX

ACPT

AUDIT IDENTITY

v2

Vyper 0.4.3

UNATTESTED

KEY EVIDENCE



Fork evidence

Route and lifecycle evidence passed at block 25,737,717.

Fuzz proof

Counterexamples were incorporated into final finding validation.

Release constraint

V17 runtime is 24,552 bytes: 24 bytes below EIP-170.

04 SCOPE

Canonical audited contracts - page 5

The final reviewed scope contains 8 production contract(s). This page lists canonical audited files only.

1 C

OK

Reviewed production contract

FINAL

2 C

OK

Reviewed production contract

FINAL

3 b

OK

Reviewed production contract

FINAL

RELEASE STATE

Release: UNATTESTED | 7 findings | 4 fixed | 3 accepted

Public source and evidence are published with this report.

04 SCOPE

Canonical audited contracts - page 6

The final reviewed scope contains 8 production contract(s). This page lists canonical audited files only.

1 b

OK

Reviewed production contract

FINAL

2 b

v17

Reviewed production contract

FINAL

3 i

OK

Reviewed production contract

FINAL

RELEASE STATE

Release: UNATTESTED | 7 findings | 4 fixed | 3 accepted

Public source and evidence are published with this report.

04 SCOPE

Canonical audited contracts - page 7

The final reviewed scope contains 8 production contract(s). This page lists canonical audited files only.

1 i

OK

Reviewed production contract

FINAL

2 i

OK

Reviewed production contract

FINAL



RELEASE STATE

Release: UNATTESTED | 7 findings | 4 fixed | 3 accepted

Public source and evidence are published with this report.

06 SECURITY CONSIDERATIONS & THREAT MODEL

Source-grounded threats and security properties

Threat modeling follows the canonical findings, limits, and methodology for Boosthub Staking v2.

Reward valuation

BH-V2-LOW-001: accepted. See Findings.

Converter protection

BH-V2-LOW-002: fixed. See Findings.

Disabled-route rescue

BH-V2-LOW-003: accepted. See Findings.

Residual reward carry

BH-V2-LOW-004: accepted. See Findings.

Rescue timer

BH-V2-LOW-005: fixed. See Findings.

sdYB classification

BH-V2-LOW-006: fixed. See Findings.

KEY SECURITY PROPERTIES USED IN LATER TESTING

BH-V2-LOW-001 ACCEPT

BH-V2-LOW-002 FIXED

BH-V2-LOW-003 ACCEPT

BH-V2-LOW-004 ACCEPT

BH-V2-LOW-005 FIXED

BH-V2-LOW-006 FIXED

TEST EVIDENCE QUALIFICATION

Terminal counterexamples informed finding validation; no fuzz run remains incomplete.

09 AUDIT METHODOLOGY & SECURITY PROCESSES

Campaign methods

Build & testing

PASS

V7

WHAT HAPPENED

Built the audited source and exercised edge cases, negative paths, invariants, and regression scenarios against executable evidence.

WHAT IT MEANS

This evidence informed the final smart-contract security verdict.

Architecture review

PASS

V7

WHAT HAPPENED

Reviewed trust boundaries, privileged roles, dependencies, state transitions, and failure paths across the declared contract scope.

WHAT IT MEANS

This evidence informed the final smart-contract security verdict.

EVIDENCE

8 contracts | 7 findings

17 public evidence files

1 pinned-fork record

10 AUDIT METHODOLOGY & SECURITY PROCESSES

Campaign methods

Economic review

PASS

V7

WHAT HAPPENED

Examined fee, reward, conversion, accounting, and incentive mechanics for leakage, manipulation, and nonlinear failures.

WHAT IT MEANS

This evidence informed the final smart-contract security verdict.

Findings validation

PASS

V7

WHAT HAPPENED

Reproduced candidates, removed unsupported hypotheses, assigned severity, and bound retained findings to evidence.

WHAT IT MEANS

This evidence informed the final smart-contract security verdict.

EVIDENCE

8

scope

7

finds

1

forks

Fuzz: COMPLETE

11 AUDIT METHODOLOGY & SECURITY PROCESSES

Campaign methods

Fork lifecycle

PASS

V7

WHAT HAPPENED

Exercised lifecycle and integration transitions on pinned fork evidence beyond isolated unit paths.

WHAT IT MEANS

This evidence informed the final smart-contract security verdict.

Manual review

PASS

V7

WHAT HAPPENED

Reviewed access control, accounting, reward flows, withdrawals, external calls, and state invariants line by line.

WHAT IT MEANS

This evidence informed the final smart-contract security verdict.

EVIDENCE

8

scope

7

finds

1

forks

Fuzz: COMPLETE

12 AUDIT METHODOLOGY & SECURITY PROCESSES

Campaign methods

Campaign evidence

PASS

V7

WHAT HAPPENED

The canonical campaign report PHASE-0-REPORT-v1.md supplied evidence used in the final contract-security review.

WHAT IT MEANS

This campaign evidence informed the final audit record.

Remediation review

PASS

V7

WHAT HAPPENED

Independently checked fixes and regression evidence, separating verified fixes from accepted bounded risks.

WHAT IT MEANS

This evidence informed the final smart-contract security verdict.

EVIDENCE

8

scope

7

finds

1

forks

Fuzz: COMPLETE

13 AUDIT METHODOLOGY & SECURITY PROCESSES

Campaign methods

Risk specification

PASS

V7

WHAT HAPPENED

Converted observed behavior into security properties, assumptions, residual risks, and release qualifications.

WHAT IT MEANS

This evidence informed the final smart-contract security verdict.

PASS

WHAT HAPPENED

WHAT IT MEANS

EVIDENCE

8

scope

7

finds

1

forks

Fuzz: COMPLETE

14 AUDIT PHASE RESULTS

Source, model, and execution results

P6

PASS

Exact-source builds and adversarial records reached terminal results.

P3

PASS

Trust boundaries, privileges, dependencies, and attack paths were reviewed.

P5

FIND

Accounting, reward, conversion, and incentive mechanics were challenged.

P8

PASS

Seven Low findings received terminal validation and disposition.

For

PASS

Pinned-fork lifecycle and route behavior were exercised.

7 Low: 4 fixed; 3 accepted.

15 AUDIT PHASE RESULTS

Dynamic evidence, validation, and release state

P4

FIND

Control flow, authority, accounting, and external calls were reviewed.

P0

PASS

Scope, source identity, and review boundaries were admitted.

P9

PASS

Four supplied fixes were independently regression-verified.

P2

PASS

Security properties, assumptions, and boundary cases were frozen.

RPT

PASS

PASS: 7 Low findings; 4 fixed/verified and 3 accepted bounded risks.

KEY QUALIFICATION

V17 has 24-byte EIP-170 headroom; changes require recompilation and review.

7 Low: 4 fixed; 3 accepted.

16 SECURITY CLAIMS & EVIDENCE

Material conclusions and their canonical evidence

01

Finding state

EVIDENCE

Open findings: C0 H0 M0 L0.

02

Remediation state

EVIDENCE

4 findings fixed and independently verified.

03

Canonical scope

EVIDENCE

8 audited contracts in normalized scope.

04

Compiler evidence

EVIDENCE

Vyper 0.4.3

05

Adversarial evidence

EVIDENCE

2 fuzz; 1 fork record(s).

06

Release state

EVIDENCE

Release state: UNATTESTED.

17 LOW - BH-V2-LOW-001

Reward-preview valuation

ACCEPT

ROOT CAUSE

A depressed valuation of accrued, unharvested rewards can give a new depositor a marginal share advantage. Fork evidence showed very small impact and no scalable extraction loop.

SOURCE TRACE

Affected: CurveYieldVault.sol

WHY THIS TRANSFERS VALUE

Accepted - bounded Low residual risk.

SIMPLIFIED ECONOMIC WITNESS

A depressed valuation of accrued, unharvested rewards can give a new depositor a marginal share advantage. Fork evidence showed very small impact and no scalable extraction loop.

SCALING, SEVERITY, AND RECOMMENDED FIX

Accepted - bounded Low residual risk.

Accepted - bounded Low residual risk.

18 LOW SEVERITY ISSUE - BH-V2-LOW-002

Converter execution protection

FIXED

ROOT CAUSE

The baseline used insufficiently independent market values for conversion protection, allowing reward-processing reverts under adverse market conditions.

FAILURE SEQUENCE

The baseline used insufficiently independent market values for conversion protection, allowing reward-processing reverts under adverse market conditions.

SOURCE TRACE

Affected: CurveYieldStakingStrategyV2.sol; converter routes

SECURITY IMPACT

Fixed / independently verified.

WHY LOW / RECOMMENDED FIX

Oracle-aware pricing and explicit execution tolerance replaced the affected converter behavior.



19 BH-V2-LOW-002 REMEDIATION

Converter protection

VERIFIED

Remediation implemented and verified

CHANGE

Oracle-aware pricing and explicit execution tolerance replaced the affected converter behavior.

FIXED / VERIFIED.

SOURCE BINDING

Status is bound to the exact audited source set.

REGRESSION

contract-findings_v1.json	PASS
FIXED / VERIFIED.	

20 LOW - BH-V2-LOW-003

Disabled-route owner rescue

ACCEPT

ROOT CAUSE

After a route is disabled and the delay passes, the owner can rescue reward-token balances. Privilege is required and deposited principal is not exposed.

SOURCE TRACE

Affected: CurveYieldStakingStrategyV2.sol

WHY THIS TRANSFERS VALUE

Accepted - bounded Low residual risk.

SIMPLIFIED ECONOMIC WITNESS

After a route is disabled and the delay passes, the owner can rescue reward-token balances. Privilege is required and deposited principal is not exposed.

SCALING, SEVERITY, AND RECOMMENDED FIX

Accepted - bounded Low residual risk.

Accepted - bounded Low residual risk.

21 LOW SEVERITY ISSUE - BH-V2-LOW-004

Residual reward carry

ACCEPT

ROOT CAUSE

Disabling a reward can remove the rescheduling path for residual reward_remaining, leaving that reward carry stranded. Deposited principal is unaffected.

FAILURE SEQUENCE

Disabling a reward can remove the rescheduling path for residual reward_remaining, leaving that reward carry stranded. Deposited principal is unaffected.

SOURCE TRACE

Affected: CurveYieldStakingStrategyV2.sol

SECURITY IMPACT

Accepted - bounded Low residual risk.

WHY LOW / RECOMMENDED FIX

Accepted - bounded Low residual risk.

22 LOW - BH-V2-LOW-005

Stale rescue timer

FIXED

ROOT CAUSE

A rescue timestamp could survive route re-enablement, letting a later disable/rescue cycle reuse an aged timer instead of a fresh delay.

SOURCE TRACE

Affected: CurveYieldStakingStrategyV2.sol

WHY THIS TRANSFERS VALUE

Fixed / independently verified.

SIMPLIFIED ECONOMIC WITNESS

A rescue timestamp could survive route re-enablement, letting a later disable/rescue cycle reuse an aged timer instead of a fresh delay.

SCALING, SEVERITY, AND RECOMMENDED FIX

Reward-route re-enablement now clears stale rescue timing before any later disable/rescue cycle.

Fixed / independently verified.



23 BH-V2-LOW-005 REMEDIATION

Rescue timer reset

VERIFIED

Remediation implemented and verified

CHANGE

Reward-route re-enablement now clears stale rescue timing before any later disable/rescue cycle.

FIXED / VERIFIED.

SOURCE BINDING

Status is bound to the exact audited source set.

REGRESSION

contract-findings_v1.json	PASS
FIXED / VERIFIED.	

24 LOW SEVERITY ISSUE - BH-V2-LOW-006

sdYB reward classification

FIXED

ROOT CAUSE

The constructor treated WANT as a required BoostHub-native reward. This blocked PID2/sdYB because sdYB is WANT but not a native reward.

FAILURE SEQUENCE

The constructor treated WANT as a required BoostHub-native reward. This blocked PID2/sdYB because sdYB is WANT but not a native reward.

SOURCE TRACE

Affected: BoostHubStaking-v17.vy

SECURITY IMPACT

Fixed / independently verified.

WHY LOW / RECOMMENDED FIX

WANT accounting is separated from BoostHub-native reward-source validation for the sdYB pool.



25 BH-V2-LOW-006 REMEDIATION

sdYB classification

VERIFIED

Remediation implemented and verified

CHANGE

WANT accounting is separated from BoostHub-native reward-source validation for the sdYB pool.

FIXED / VERIFIED.

SOURCE BINDING

Status is bound to the exact audited source set.

REGRESSION

contract-findings_v1.json	PASS
FIXED / VERIFIED.	

26 LOW - BH-V2-RR2-LOW-001

Fee-conversion minOut tolerance

FIXED

ROOT CAUSE

The first fix passed an oracle preview as exact minOut. Normal quote/execution divergence could revert fee conversion without a security-significant anomaly.

SOURCE TRACE

Affected: BoostHubStaking-v17.vy

WHY THIS TRANSFERS VALUE

Fixed / independently verified.

SIMPLIFIED ECONOMIC WITNESS

The first fix passed an oracle preview as exact minOut. Normal quote/execution divergence could revert fee conversion without a security-significant anomaly.

SCALING, SEVERITY, AND RECOMMENDED FIX

V17 adds 1% default staking conversion tolerance, admin configurability, and a 4% hard maximum.

Fixed / independently verified.



27 BH-V2-RR2-LOW-001 REMEDIATION

Fee minOut tolerance

VERIFIED

Remediation implemented and verified

CHANGE

V17 adds 1% default staking conversion tolerance, admin configurability, and a 4% hard maximum.

FIXED / VERIFIED.

SOURCE BINDING

Status is bound to the exact audited source set.

REGRESSION

contract-findings_v1.json	PASS
FIXED / VERIFIED.	

28 KEY EVIDENCE INDEX

Public source identity, test evidence, and release records

PUBLIC RELEASE

CurveYield2/Audits Boosthub Staking v2 v2
CurveYieldGovernanceToken.sol
CurveYieldVault.sol
CurveYieldStakingStrategyV2.sol

PUBLIC EVIDENCE FILES

FILE / ROLE / STATUS
E1 smart-contract-scope_v1.json / published
E2 contract-findings_v1.json / published
E3 contract-methodology-summary_v1.json / published
E4 CurveYieldGovernanceToken.sol / published
E5 CurveYieldVault.sol / published
E6 CurveYieldStakingStrategyV2.sol / published

Selected evidence is client-safe and linked through the stable public release base. Total public evidence files: 17.

29 KEY EVIDENCE INDEX

Additional public evidence and publication state

ADDITIONAL PUBLIC EVIDENCE

FILE / ROLE / STATUS
E1 SdYbRewardConverter.sol / published
E2 BoostHubStaking-v17.vy / published
E3 ICurveChildGauge.sol / published
E4 ICurveYieldVault.sol / published
E5 ICurveYieldVaultStrategy.sol / published
E6 MEDUSA-TERMINAL-RECORD-v3.json / published

PUBLICATION AND RELEASE STATE

Base URL Audits	Files 17	Contracts 8
QA status PASS	Release UNATTESTED	PDF v2

30 KEY EVIDENCE INDEX

Public source identity, test evidence, and release records

PUBLIC RELEASE

CurveYield2/Audits Boosthub Staking v2 v2
CurveYieldGovernanceToken.sol
CurveYieldVault.sol
CurveYieldStakingStrategyV2.sol

PUBLIC EVIDENCE FILES

FILE / ROLE / STATUS
E1 NATIVE-FUZZ-TERMINAL-RECORD-v3.json / published
E2 PHASE6_DETERMINISTIC_RESULTS-v1.json / published
E3 solidity-build-summary-v1.json / published
E4 vyper-build-summary-v1.json / published
E5 PHASE6_MEDUSA_CONFIG-v1.json / published

Selected evidence is client-safe and linked through the stable public release base. Total public evidence files: 17.

31 CONCLUSION

Final assessment of the exact Boosthub Staking v2 audited source suite

FINAL DISPOSITION

PASS

C0 H0 M0 L0

RELEASE STATUS

UNATTESTED

7 Low findings: 4 fixed/verified; 3 accepted bounded residual risks.

WHAT THE AUDIT ESTABLISHED

The normalized audit model contains 8 audited contract(s), 7 canonical finding(s), and 17 public evidence file(s).

FINAL SOURCE IDENTITY

Public release bundle

8 audited contracts; exact identities are in Scope.

DEPLOYMENT QUALIFICATION

Deployment qualification: UNATTESTED. This report makes no deployment claim beyond the canonical release evidence.

IMPORTANT REMAINING LIMITATION

V17 has 24 bytes of EIP-170 headroom; bytecode-affecting changes require recompilation and review.

FINAL RUNTIME SIZES

BoostHubStaking-v17.vy: 24,552 bytes | EIP-170 limit: 24,576 bytes | headroom: 24 bytes